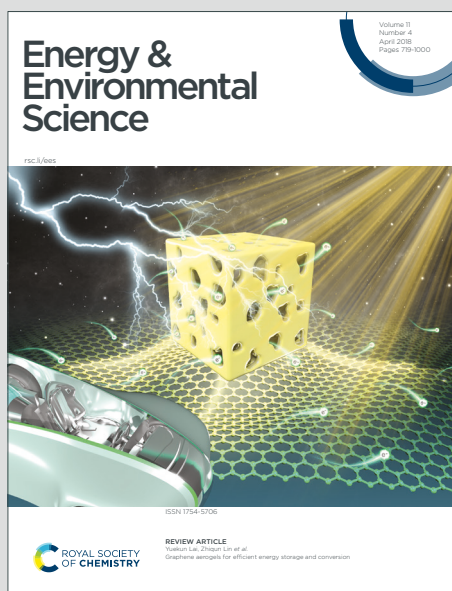


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Broader context

In recent years, lead (Pb)-based perovskite solar cells (PSCs) are of high interest because of their excellent photovoltaic performance. However, the presence of Pb in perovskite materials also gives rise to a range of significant Pb-related issues throughout the entire life cycle of PSCs. These issues include the generation of Pb-related defects during preparation, device degradation due to the photolysis of excessive PbI_2 during operation, and the potential risk of Pb^{2+} contamination when devices are damaged. These Pb-related issues severely impede the commercialization of PSCs. However, currently, there is a lack of systematic and targeted solutions. Herein, a novel yet facile Pb-management strategy has been proposed to alleviate the Pb-related issues by incorporating benzophenone (BP) additives, namely BP-5 and BP-9, in the perovskite layer. Results have confirmed the efficacy of BP additives to achieve the highly efficient, stable and environmentally-friendly PSCs by passivating Pb-related defects, suppressing the formation of δ -FA phase, sustainably absorbing the ultraviolet (UV) and forming water-insoluble complexes with Pb^{2+} . This study presents a key perspective on the further development of PSCs from addressing Pb-related issues throughout the entire life cycle of PSCs.

Lead (Pb) Management in the Entire Life Cycle of Highly Efficient and Stable Perovskite Solar Cells

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Abstract

Lead (Pb) related issues severely affect the entire life-cycle of perovskite solar cells (PSCs) which includes Pb-related defects during preparation, photolysis of excessive PbI_2 during the operation and Pb^{2+} leakage from the damaged PSCs towards the end of lifecycle, consequently, hindering the commercialization of PSCs. Herein, a novel yet facile Pb-management strategy has been proposed to alleviate the Pb-related issues by incorporating benzophenone (BP) additives, namely BP-5 and BP-9, in the perovskite layer. The resultant PSCs showed excellent efficiency (25.47%) and stability by successfully passivating Pb-related defects, suppressing the formation of δ -FA phase, sustainably absorbing the ultraviolet (UV) and forming water-insoluble Pb^{2+} complexes to limit the Pb leakage. The devices retained 96% and 91.15% of their initial efficiency after 600 h and 1000 h of UV aging test and operational test, respectively. Moreover, the addition of BP-9 effectively suppressed Pb^{2+} leakage by 93.5%. Despite its simple approach, the proposed strategy has successfully addressed Pb related issues spanning over the entire life-cycle of the PSCs.

Keywords: Pb-related defects, UV-stability, Pb^{2+} leakage, entire life-cycle Pb management

Introduction

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Lead (Pb)-based PSCs have drawn an unprecedented attention from researchers due to their skyrocketing efficiency increasing from 3.8%¹ in 2009 to over 26%² in 2023. Moreover, owing to the multiple advantages of perovskite materials which include outstanding photoelectric properties, cost-effective fabrication, light-weight and the thin-film nature, PSCs are expected to be the strongest market competitor for the silicon solar cells. However, given the instability and toxicity of Pb-based perovskite, the road to commercialization of PSCs remains arduous.³⁻⁶

Intensive studies^{5, 7, 8} have demonstrated that the existence of Pb element in perovskite materials despite of its benefits, possesses harm to the performance of PSCs. For instance, existence of Pb in perovskite causes a strong spin-orbit coupling effect in the electronic band structure, contributing to smaller effective mass of electrons.^{7, 9, 10} Thus, Pb-based perovskites exhibit extraordinary diffusion coefficients and mobilities,⁷ contributing to the transport of charge carriers and eventually enhancing the efficiency of PSCs. On the other hand, existence of Pb in perovskites also brings a series of serious and undesirable consequences. Firstly, undercoordinated Pb²⁺ on the surface of perovskite film^{11, 12} might appear due to the over-annealing and the tensile stress¹³ during the preparation of perovskite. These Pb-related defects usually serve as deep-level trap states in perovskite,^{13, 14} inevitably triggering severe non-radiative recombination in perovskite films and lower open-circuit voltage (V_{oc}) of PSCs.¹⁵ Additionally, Chen et al.¹⁶ have reported that the undercoordinated Pb²⁺ at grain boundaries of perovskite contributes to the formation of δ -FAPbI₃ and the generation of non-active perovskite phase can be successfully inhibited by incorporating chemical chelation with undercoordinated Pb²⁺. As mentioned above, effective passivation of Pb-related deep-level defects enhances the efficiency of PSCs and addresses their performance constraints. Secondly, the operational stability of PSCs is also significantly limited by the Pb-related issues. Owing to the advantage of PbI₂ passivation effect in perovskite films,¹⁷ many processes of highly efficient PSCs are developed to incorporate over-stoichiometric ratio of PbI₂,¹⁸⁻²⁰ especially the devices

fabricated by two-step sequential spin-coating method.^{3, 20} However, the excessive PbI_2 and undercoordinated Pb^{2+} can easily decompose into metallic Pb under the stimuli of UV light in the solar spectrum during the operation of PSCs,^{3, 8, 21} thus triggering the degradation of devices during the operation.²¹ In this context, some previous studies employed the UV filter to cut the UV wavelength in order to protect PSCs when devices were tested at maximum power point (MPP).^{22, 23} Additionally, many reported operational stabilities of PSCs were tested with white light-emitting diode (LED) as light source,²⁴⁻²⁶ while the white LED usually does not contain UV wavelength. Considering the inevitable influence of UV light in real operational conditions, an in-depth exploration of the impact of UV light on the operational stability of perovskite solar cells is imperative. Consequently, the implementation of effective mitigation strategies to enhance the UV stability of PSCs becomes necessary. Thirdly, although the effective physical encapsulation can reduce Pb leakage, the issue of Pb leakage from PSCs remains a considerable concern, particularly due to prolonged exposure of the panels to the uncontrolled atmospheric conditions throughout their operational lifespan. In the event of potential damage to the panels, the dissolution and dispersion of Pb as a consequence of rainfall can also occur.^{5, 6} Moreover, the release of toxic Pb^{2+} from broken devices at the end of the life-cycle of PSCs is of the prominent concern, potentially prompting stringent legislation regarding their usage. Normally, despite the relative low Pb concentrations in complete photovoltaic modules, the unit area concentration of Pb in perovskite films is assessed to be about 0.75 g m^{-2} , which is over 100 times higher than that of present day Pb-containing paints.^{5, 6} Hence, the potential risk of Pb^{2+} leakage and their enormous hazard for environment are serious issues which need immediate and serious attention. .

As mentioned above, the Pb-related issues affect the entire life-cycle of PSCs and severely hinder the true commercial application of PSCs. However, to date, it is still challenging to realize systematical management of Pb-related issues throughout the life-cycle of PSCs, including passivation of Pb-related defects, inhibition of the photolysis of PbI_2 under stimuli of UV light and alleviating the Pb^{2+} leakage. Therefore,

a comprehensive strategy is urgently in need to solve the Pb-related problems for highly efficient, stable and environmentally-friendly PSCs.

Herein, we successfully demonstrated a very effective yet facile Pb management strategy covering the entire life cycle of highly efficient and stable PSCs by introducing BP additives (BP-5, BP-9). Experimental evidence and corresponding characterizations have proved that the functional groups (SO_3^-) in BP additives interact with Pb^{2+} to simultaneously regulate the crystallization of perovskite to inhibit the formation of $\delta\text{-FAPbI}_3$ and passivate the Pb-related deep-level traps of perovskite films, which eventually contributed to the highly efficient PSCs with 25.47%. Additionally, the target device exhibited a high external quantum efficiency (EQE_{EL}) of 17.88% when operated as light emitting diode (LED). To the best of our knowledge, the EQE_{EL} value is one of the highest reported values among the devices fabricated by two-step sequential method.^{20, 27, 28} Moreover, due to the sustainable absorption of UV light by BP-5 molecules, target PSCs have exhibited excellent UV stability and devices maintained 96% of initial efficiency after 620 h of continuous operation under UV light (22 mW/cm^2). In comparison, the control device exhibited $\sim 30\%$ loss in the initial PCE only after ~ 100 h. Finally, to inhibit the leakage of Pb, we incorporated BP-9 in perovskite films, since the BP-9 molecule possesses the double chelating sites with Pb^{2+} . In principle, it can bind more strongly with perovskite than BP-5, forming the insoluble complexes with Pb^{2+} in water. Eventually, we mixed the two kinds of BP additives (BP-5, BP-9) and incorporated in PSCs. Experimental results exhibited that after incorporation of BP-9 additives, the Pb leakage of PSCs has been effectively suppressed without sacrificing the photovoltaic performance of devices. This study presents a straightforward yet promising approach aiming at resolving the Pb-related issues covering the entire life cycle of PSCs, thereby paving the way towards highly efficient and stable perovskite solar cells.

Results and Discussion

To solve Pb-related issues in PSCs, we proposed the entire life cycle Pb management strategy by incorporating the BP additives in perovskite films. As shown in Figure 1a, the BP-5 additive is best known for UV absorbent capability. From here onwards, we denote the reference perovskite as 'Control' and the BP additives containing perovskite as 'Target'. X-ray photoelectron spectroscopy (XPS) analysis was conducted to investigate the interaction between BP-5 additives and perovskite materials. As depicted in Figure 1b, the significant signals of 'S' element were detected only in target films, indicating successful introduction of BP-5 additives. Interestingly, after incorporating BP-5 additives, XPS spectra of the Pb 4f orbital peak showed a significant deviation towards lower binding energy (Figure 1c), indicative of the changed surrounding electron cloud density, which is probably due to the interaction between undercoordinated Pb^{2+} in perovskite and SO_3^- group in BP-5 additives.²⁹ Moreover, compared to the control film, the metallic Pb in target film is obviously decreased. Since undercoordinated Pb^{2+} has usually been considered as the main factor leading to the generation of metallic lead,^{13, 30} the less metallic Pb in target film further demonstrates the effective interaction between BP-5 additives and undercoordinated Pb^{2+} . Generally, as mentioned above, undercoordinated Pb^{2+} and metallic Pb serve as deep traps that can cause severe non-radiative recombination, thus detrimentally affecting the efficiency and stability of PSCs. These XPS analysis results preliminarily verified the positive role of BP-5 additives in the management of Pb-related defect state density.

To directly verify the interaction between Pb^{2+} and BP-5 additives, we then tested the crystallinity and absorbance spectra of PbI_2 film. Figure 1d depicted that after incorporating BP-5 additives, a new peak appeared at 9.75° and the intensity of the new peak increased with enhanced concentration of additives, while the peak of PbI_2 showed obvious decline with the increase of additives in PbI_2 film. These results indicate the generation of a new product due to the interaction between the PbI_2 and BP-5 additives. Moreover, to exclude the possibility that a new peak is derived from the additive itself,

as shown in Figure 1e, we further tested the additives in the form of film and powder. The results clearly exhibited that the new peak at 9.75° only exists in PbI_2 films (incorporated with BP-5), indicating the new peak is not derived from the additives itself but as the reaction product of PbI_2 and BP-5 additives. Additionally, UV-vis absorbance spectra of PbI_2 in Figure 1f exhibited that the bandgap of PbI_2 films increased from 2.26 eV to 2.73 eV with the increase of incorporated BP-5 additives. The increased bandgap of PbI_2 films is probably associated with the generation of new products, as evidenced by XRD test results.

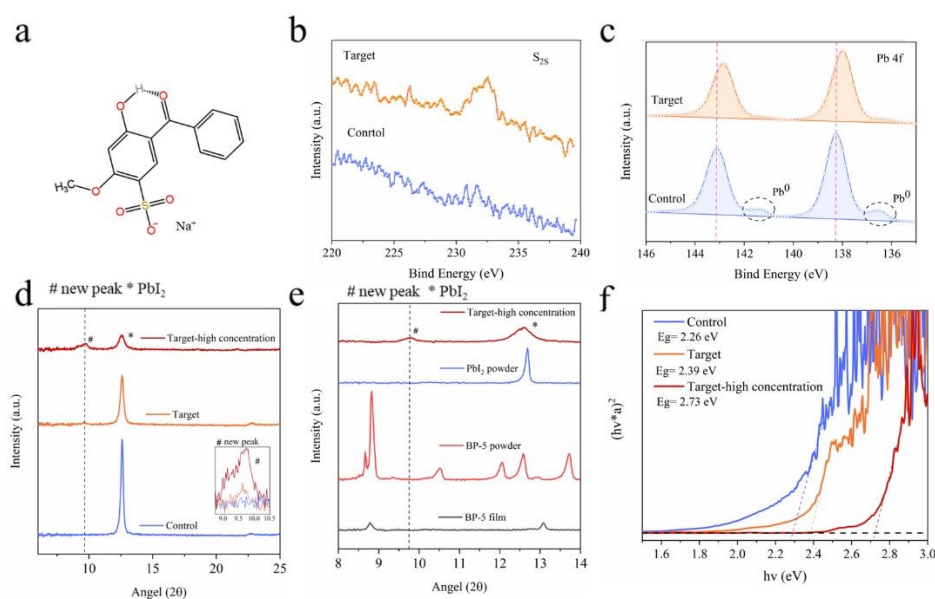


Figure 1 The interaction between BP-5 additives and perovskite materials. (a) The chemical structure of BP-5 additive. (b-c) XPS analysis of control and target-perovskite films. (b) deconvoluted XPS spectra for S 2s; (c) deconvoluted XPS spectra for Pb 4f. (d) XRD spectra of PbI_2 films with different concentration of BP-5 additives (0, 0.3%, 1%). (e) XRD spectra of BP-5 (powder and films) and PbI_2 (powder and films). (f) UV-vis absorbance spectra of PbI_2 films with different concentration of BP-5 additives (0, 0.3%, 1%)

After verifying the strong interaction between undercoordinated Pb^{2+} in perovskite and SO_3^- group in BP-5 additives, GIWAXS and SEM were carried out to further

explore the role of BP-5 additives on crystallization and morphology of perovskite. As shown in Figure 2a, 2D-GIWAXS patterns of control film exhibit an obvious ring at 8.3 nm^{-1} , which is considered as the diffraction peak of δ -FAPbI₃.³¹ However, as depicted in Figure 2b, there is no such ring in target film, showing that BP-5 additives can effectively suppress the formation of δ -FAPbI₃. To further elucidate, we tested GIWAXS of perovskite at various incident angles ranging from 0.1° to 0.6° . As depicted in Figure S1 and Figure 2c-d, the signal of δ -FAPbI₃ is not found at different incident angles of 2D-GIWAXS, which indicates that there is no δ -FAPbI₃ either on surface or bulk of the target film. This result further proved the vital role of BP additives in suppression of δ -FAPbI₃. Moreover, the first-principles calculations were performed to investigate the effect of BP-5 on crystallization of FAPbI₃ (Figure S2). DFT calculation results revealed that the total energy of δ -FAPbI₃ (-4.633 eV/atom) is smaller than that of α -FAPbI₃ (-4.619 eV/atom), indicating that δ -FAPbI₃ is more stable than α -FAPbI₃ at room temperature. This result is consistent with previous studies. However, after incorporation of BP-5 molecule on the surface of δ -FAPbI₃ and α -FAPbI₃, respectively, the energy of α -FAPbI₃ is smaller than that of δ -FAPbI₃. As depicted in Figure S2, from the perspective of energy, the δ -FAPbI₃ with the BP-5 molecule tends to transform to α -FAPbI₃. The calculation result agrees with the beforementioned experimental work that the δ -FAPbI₃ phase disappeared in target films (2D-GIWAXS, Figure 2a-d). Moreover, the integrated 1D-XRD spectra of perovskite in Figure 2c has demonstrated that BP-5 additives contribute to enhancing the crystallinity of perovskite. Figure 2c shows that FWHM of the target film has decreased compared to the control film. Additionally, the top view SEM images of perovskite films in Figure 2e-f show that in comparison to control film, the grain size of target perovskite film significantly increases, which agrees well with the decreased FWHM of target perovskite in Figure 2d. As mentioned above, BP-5 additives can interact with the perovskite, which slows down the crystallization and reduces the number of nuclei, thereby leading to the enlarged grains in perovskite films. In general, the larger crystal grains of perovskite films indicate the less grain boundaries in PSCs, which contributes to the photovoltaic

performance of PSCs. Moreover, we have employed the UPS to investigate the influence of BP additives on the work function of perovskite films. The results as shown in Figure S3 have revealed that after incorporating the BP additives, the work function and fermi level of target perovskite films show downshift. Generally, previous literatures³²⁻³⁴ have demonstrated that the excess PbI_2 exists in perovskite films lead to the generation of the n-type perovskite films due to the self-doped effect. Thus, the downshift of fermi level of target perovskite films could be attributed to the less residual PbI_2 in films (as proofed by GIWAXS results in Figure 2d) Additionally, the interaction between BP additives and Pb^{2+} contributes to retarding the crystallization process of perovskite. Delaying the crystallization process is conducive to allowing PbI_2 to have sufficient reaction time with organic salts, thereby reducing residual PbI_2 .^{35, 36}

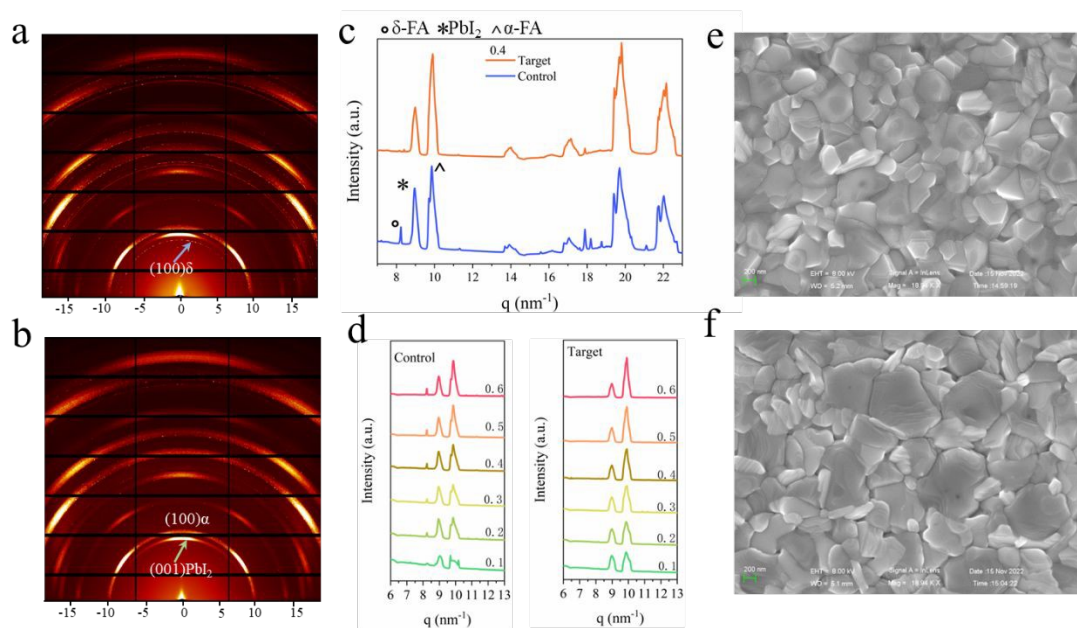


Figure 2. The crystallinity and morphology of control and target perovskite films. (a, b) GIWAXS images of control (a) and target perovskite films (b). (c) The integrated 1D-XRD spectra of perovskite films. (d) The integrated 1D-XRD spectra of perovskite at various incident angel ranging from 0.1° to 0.6° . (e, f) SEM images of control (e) and target perovskite films (f).

After achieving the remarkable crystallinity and morphology of perovskite films, we then employed the two-step sequential spin-coating method to fabricate devices with the configuration of *ITO/SnO₂/Perovskite/PEAI/Spiro-OMeTAD/Au* as shown in Figure 3a. The concentration of BP-5 additives incorporated in perovskite has been optimized carefully and corresponding photovoltaic performance of the target PSCs and control PSCs has been presented in Figure S4. Target devices with 0.3% BP-5 additives showed the best performance and achieved an average efficiency of 24.64% (Figure 3b). Moreover, as depicted in Figure 3c, compared to the champion device of control PSCs, the efficiency of champion target device (0.3% BP-5) is greatly enhanced to 25.13% with a V_{oc} of 1.1913V, FF of 82.25% and a J_{sc} of 25.65 mA/cm². Moreover, compared with control devices, the hysteresis index of target devices has significantly decreased. EQE plots depicted in Figure 3d shows that integrated J_{sc} (25.22 mA/cm²) obtained from EQE matches well with that of extracted from J - V curves (25.65 mA/cm²) and the deviation is only 1.68%. In addition, to verify the efficiency of target devices, one of the target devices was validated by an independent solar cell-accreditation laboratory (SIMIT, Shanghai, China) for certification, in which a PCE of 24.42% has been certified, as shown in Figure S5. Target device was held at a bias of 1.04V for nearly 300s, leading to a higher and more stable power output (SPO) compared with control device (Figure 3e). As depicted in Figure S4 and Figure 3c, compared with control devices, the greatly enhanced efficiency of target device is ascribed to the higher V_{oc} and FF, which are likely attributed to less Pb-related deep-level defects and high crystallinity (Figure 2c). Since an efficient solar cell is usually an efficient LED, we therefore tested the solar cell device as an LED in the dark and under a voltage bias to estimate the recombination in devices (Figure 3f). Target devices showed 17.88% EL efficiency under an injection current of 25.42 mA cm⁻², while the control device showed 3.56% EL efficiency under almost same injection current. The EQE_{EL} value of target device is one of the highest reported value in PSCs to the best of our knowledge.^{20, 27, 28} We then calculated the V_{oc} loss for non-radiative recombination (ΔV_{oc_n}) according to value of EQE_{EL},³⁷ as shown in the inset figure. ΔV_{oc_n} of the target device and control

device is 45 mV and 87 mV, respectively, which means that the V_{oc} loss for non-radiative recombination has been significantly suppressed. Moreover, we also calculated the theoretical value of V_{oc} ($\Delta V_{oc-improved}$) improvement according to the tested value of EQE_{EL}^{28} . The value of $\Delta V_{oc-improved}$ is 42 mV, which is consistent with the increased value of V_{oc} depicted in Figure S4.

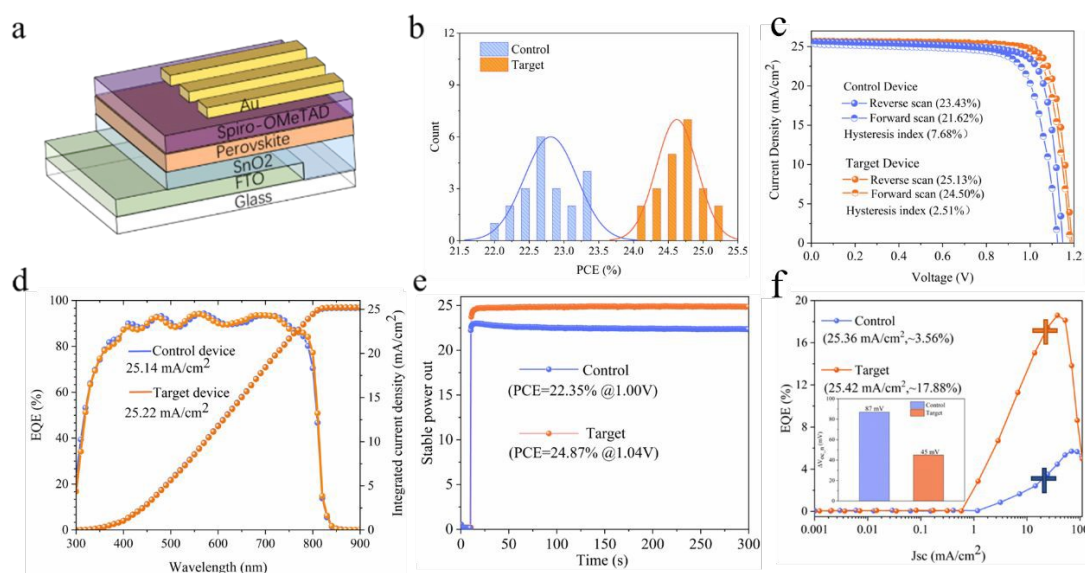


Figure 3. Photovoltaic performance of devices. (a) The configuration image of PSCs. (b) PCE statistics of the control and target devices. (c) Forward scan and reverse scan efficiencies of the champion control and target device. (d) EQE spectra and integrated current density of control and target device. (e) Stable power out (SPO) of control and target device. (f) EQE of EL of the devices while operating as LEDs and $\Delta V_{oc,n}$ of devices are shown in the inset figure.

As the significantly enhanced V_{oc} and reduced hysteresis of target devices can be witnessed, we carried out a series of experiments to systematically investigate the role of BP-5 additives in passivating the defects in PSCs. Figure 4a illustrates that the target film exhibits stronger PL intensity than that of the control film, implying the successfully inhibited trap density in target film³⁸. Moreover, the effect of BP-5 additives on film uniformity was elucidated by the 2D PL mapping imaging as viewed

in Figure S6. For the control perovskite film, the uneven and weak PL distribution implies the poor crystalline quality of film and relatively severe non-radiative recombination. However, after the addition of BP-5 additives, much more homogeneous and brighter PL emissions are observed in the target film, which confirms the significantly improved crystallinity and reduced non-radiative recombination of the sample, agreeing well with our GIWAXS results. Additionally, the TRPL measurements were conducted, and the results (Figure 4b) show that the target-perovskite sample exhibits a significant longer carrier lifetime (1.05 μs) than that of the control (0.44 μs). It is consistent with the steady-state PL measurement (Figure 4a) and PL mapping (Figure S6), wherein the enhanced PL intensity and PL decay lifetime were observed. It indicates that the nonradiative recombination of photogenerated carriers has been largely suppressed by BP-5 additives. In addition, V_{oc} of devices under various light intensities were recorded (Figure 4c). It shows that the control device showed a slope of $1.97 k_B T/q$, while target exhibited a much smaller slope of $1.69 k_B T/q$, where k_B is the Boltzmann constant, T is temperature. Given the fact that the slope of deviation from unity $k_B T/q$ usually reflects the trap-assisted non-radiative recombination in PSCs,²⁸ the smaller slope of target devices indicates that the non-radiative recombination is successfully suppressed in target PSCs. Furthermore, Mott-Schottky analyses of PSCs was conducted under a frequency of 10 kHz with bias voltages ranging from 0 to 1.2 V (Figure 4d). It is depicted that the built-in potential (V_{bi}) increased from 0.89 V for control device to 1.02 V for target PSC, which is in good agreement with the increased V_{oc} of target PSCs extracted from JV curves. Similarly, the dark current and EIS testing of devices (Figure S7-S8) are also consistent with above results. These characterizations have further demonstrated the effective and vital role of BP-5 additives to inhibit the non-radiative recombination in PSCs, which obviously contributes to the high efficiency of target devices. Generally, the non-radiative recombination is related to the trap density in perovskite films,^{28, 39} implying that the defects in perovskite films can be effectually passivated by BP-5 additives. We therefore carried out the space charge limited current (SCLC) characterization to

directly investigate the variation of trap density in perovskite films. The electron-only device with structure of ITO/SnO₂/Perovskite/ [6,6]-phenyl-C61-butyric acid methyl ester (PCBM) /Au was fabricated, and results were shown in (Figure 4e). Trap-filled voltage (V_{TFL}) of electron-only device is significantly reduced from 0.25 V to 0.18 V after incorporating BP-5 additives. Since V_{TFL} is proportional to the trap density of perovskite^{40, 41}, the lower V_{TFL} of target devices verifies that there are less defects in target perovskite films.

To further verify the positive influence of BP-5 additives on Pb-related deep-level traps, thermal admittance spectroscopy (TAS) measurement was employed to obtain more comprehensive information on the defect variation. The distribution of the defect density of states in energy space was investigated by using the following equation based on TAS characterization:

$$N_t(E_f) = -\frac{V_{bi}dC}{eWdfk_BT}f \quad (1)$$

Where N_t is the trap density of states; E_f is the energetic demarcation corresponding to each frequency. V_{bi} can be obtained from Figure 4d. C is the capacitance corresponding to each frequency; and W represents depletion width; f is the frequency. The results are presented in Figure 4f, the defect density of states in energy space can be divided into three regions: trap I, trap II and trap III. Interestingly, the density of trap I and trap II of control devices and target devices are almost the same, but the trap III intensity of target film have been significantly decreased in comparison to that of control films. Moreover, previous work illustrated that the defects in trap III are involved in deep-level traps,⁴² triggering the severe non-radiative recombination in PSCs. Hence, the above results have revealed that the inhibited non-radiative recombination in target devices is due to the reduced trap density of states and the most of decreased traps are involved in deep-level defects. This further demonstrated that BP-5 can effectively passivate those deep-level defects in PSCs. Normally, Pb-related traps act as deep-level defects in perovskite^{11, 13, 43}. Moreover, the XPS and UV-vis absorbance spectra have already revealed that BP-5 can interact with the undercoordinated Pb²⁺, thereby demonstrating

the positive role of BP-5 additives in passivating the Pb-related deep-level traps to achieve highly efficient PSCs.

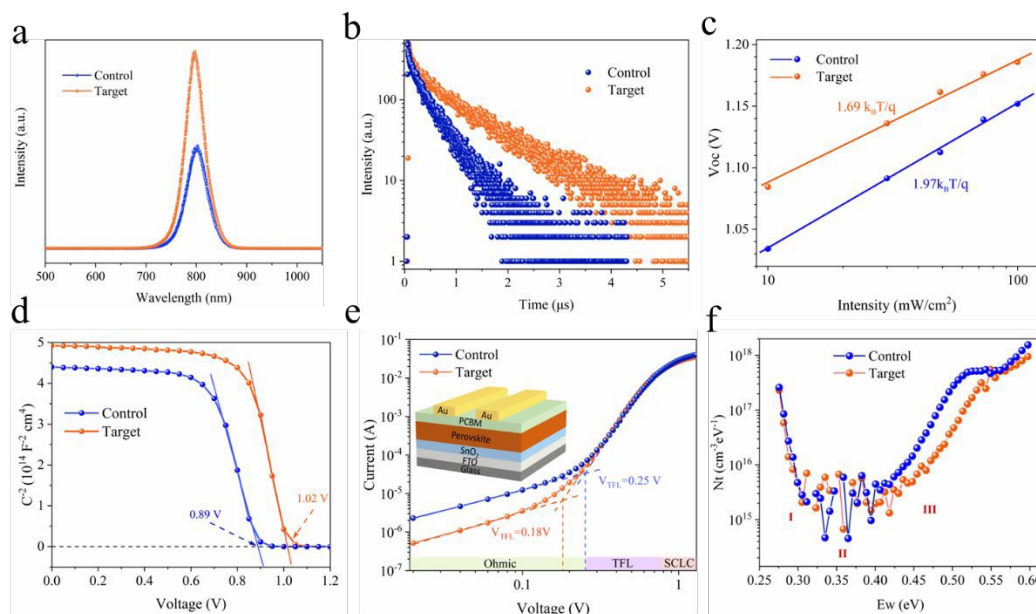


Figure 4. Characterization of perovskite films and devices. (a) PL spectra of control and target perovskite films deposited on glass substrate. (b) The TRPL testing of control and target films (perovskite/ glass). (c) V_{oc} versus light intensity for the control and target devices. (d) Mott-Schottky plot, comparison of the control and target devices. (e) SCLC curves of control and target electron-only devices. (f) Trap-state densities of the corresponding PSCs.

Notably, the high efficiency of PSCs alone is not enough to realize their commercialization due to the notorious and poor operational stability of PSCs. Additionally, owing to the positive role of excessive PbI_2 in passivating the defects, some excessive PbI_2 usually exists in the grain boundaries of highly efficient PSCs.^{17, 18} However, previous studies^{3, 21} have demonstrated that the residual PbI_2 on perovskite films is detrimental to the operational stability of PSCs because of the harmful byproduct (metallic Pb) of the photolysis of PbI_2 under the illumination of high energy

light. Additionally, since the excessive PbI_2 and undercoordinated Pb^{2+} are sensitive to UV light and can easily degrade into metallic Pb,^{11, 21} some previous studies therefore used the UV filter to filter out the UV wavelength of the solar spectrum or chose white LED as light source for protecting the PSCs from UV light when it operated at MPP.²²⁻²⁵ However, because of the non-negligible influence of UV light on PSCs, especially when PSCs works outdoors, it is of great importance to investigate the long-term UV operational stability of PSCs. We therefore carried out the related experiments to verify effectiveness of our strategy on enhancing the operational durability of PSCs under the illumination of UV light. The illustration in Figure S9 presents the UV screen mechanism of our strategy: BP-5 additives as a well-known UV absorber contains the adjacent oxygen on carbonyl and the hydrogen on hydroxyl, which can form intramolecular hydrogen bond and a chelating ring in the diphenylketone structure. The BP additives can absorb the UV light to open the chelating ring. The UV light has been consumed and the enol structure is formed, achieving the capabilities of sunscreen, thereby contributing to protecting the perovskite from the destruction of UV light. Given that PbI_2 can be decomposed into metallic Pb under stimuli of UV light, we firstly investigated the influence of BP-5 additives on inhibiting the photolysis of PbI_2 when it was illuminated under UV light. The results presented in Figure S10 illustrated that after 5 hours of illuminating at UV light (441 mW/cm²), the color of control PbI_2 film turned into black, implying the generation of metallic Pb due to the photolysis of PbI_2 .²¹ Whereas, with the increasement of BP additives in PbI_2 , less color change of PbI_2 during the illumination of UV light has been observed, which indicates that BP-5 additives can effectively inhibit the photolysis of PbI_2 by absorbing the UV light. Additionally, XPS analysis has been employed to explore the influence of UV light on perovskite films. As depicted in Figure S11, the control films shows much stronger signals of metallic Pb after the UV illumination, which indicates that the Pb^{2+} in perovskite is reduced by UV light and eventually decomposed into metallic Pb.²¹ However, negligible metallic Pb signal appeared in target films regardless of their exposure to UV light, which could be attributed to the effective passivation of under-

coordinated Pb^{2+} defects and absorption of UV light by BP-5 additives. Moreover, since the metallic Pb usually acts as the deep-level traps inducing severe non-radiative recombination in PSCs, PL mapping was conducted to further elucidate the harmful impact of UV light on the carrier recombination of perovskite films. Figure 5a-f illustrate that compared to the fresh sample, after UV illumination, the PL mapping of the control film became darker and the fluorescence intensity (average value) distribution diagram of aged control film decreases to $\sim 71\%$ of its original value. The significant reduction in the PL peak intensity of the control film after UV illumination is another evidence of the generated metallic Pb by UV light, whereas the target films exhibited higher intensity and less PL intensity change than control films. These results mean that introduced BP-5 additives in perovskite films work effectively to inhibit the decomposition of perovskite films under the UV light. Thus, based on these results, we further evaluated the effect of this strategy on operational stability of PSCs when devices work under UV light. The target devices have shown remarkable operational stability, retaining nearly 96% of initial efficiency even after ~ 620 h of MPP tracking and continuous illumination under 22 mW/cm^2 UV light (Figure 5g). Whereas, the control devices decayed to 72% of their initial PCE only after 120 h of aging test. Hence, target devices have been proved to be highly UV stable and efficient as compared to their control counterparts, which further demonstrated the success of the BP-5 additives. Additionally, the operational test of target devices under LED light has been carried out following the international organization for standardization (ISOS) testing protocols (ISOS-L-1). The target devices have shown remarkable operational stability by retaining 91.2% of the initial efficiency even after 1,000 h maximum power point tracking (MPPT) and continuous illumination as shown in Figure 5h, whereas the control device decayed to 78% of its initial PCE only after ~ 750 h of the aging test. Previous study has demonstrated that the δ -phase perovskite serves as phase impurities in perovskite films, which provides the starting points for photochemical degradation.⁴⁴ Therefore, the enhanced operational stability of target device is likely due to the BP-5 additives which removed the hexagonal phases in perovskite (Figure 2c). Moreover,

the passivation of Pb-related deep-level traps via BP-5 contributes to increasing the photo stability of PSCs.⁴³

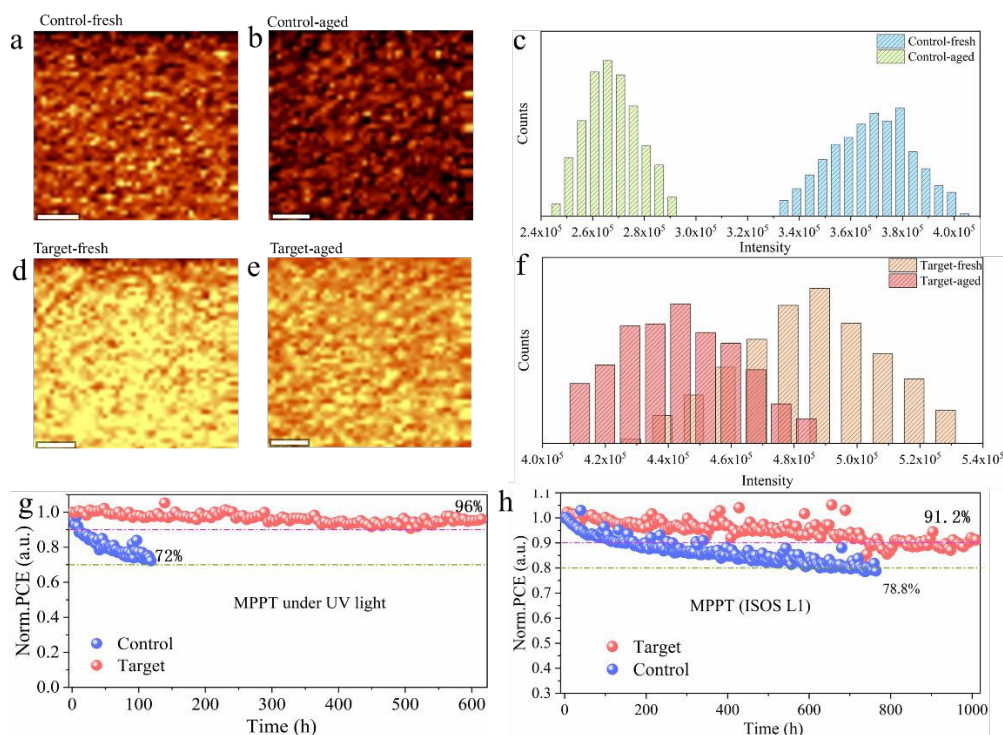


Figure 5. The stability characterization of perovskite films and devices. (a-c) PL mapping characterization of control perovskite films before (a) and after UV aging period (b). (c) The corresponding PL intensity statistics of the control perovskite films before and after UV aging period. (d-f) PL mapping characterization of target perovskite films before (d) and after UV aging period (e). (f) The corresponding PL intensity statistics of the target perovskite films before and after UV aging period. (g-h) The operational stability of devices under UV light (g) and white LED (h).

For commercialization of PSCs, the risk of Pb leakage as the last but not least Pb-related issues should be carefully taken into consideration, since the generation of Pb-related deep-level defects during fabrication and the photolysis of PbI_2 under high energy light have been successfully alleviated in target devices as proved by experimental results. Indeed, strategies for glass encapsulation can effectively reduce Pb leakage under mild environment damage, but the inhibition of Pb leakage when the encapsulation layer is broken and the water ingresses into the perovskite layer are yet

to be addressed. Therefore, the strong bonding of additives with Pb^{2+} in perovskite to form the water insoluble compound is an effective method to suppress the Pb^{2+} leakage when water ingresses into the perovskite layer. However, as shown in Figure S12, despite that the BP-5 additives can bind with Pb^{2+} in perovskite, their reaction product still easily dissolves in water, which means that the sole BP-5 additives in perovskite films cannot effectively alleviate the issue of the Pb^{2+} leakage. Since the BP-9 possesses the double sites to interact with Pb^{2+} , it may exhibit stronger interaction with Pb^{2+} and form insoluble-water complexes with the Pb^{2+} (Figure 6a). Additionally, as shown in Figure S13 and Figure S14, the result of TRPL and in-situ absorption spectra have revealed that the BP-9 can effectively passivate the defects in perovskite films and regulate the crystallization of perovskites. The results as depicted in Figure S15 have demonstrated that BP-9 contribute to enhancing the efficiency of devices. Moreover, Figure 6b illustrates that 2-terminal BP additives (BP-9) can strongly bind with Pb^{2+} and eventually generates insoluble materials in water. To further evaluate the Pb-absorbing capability of BP-9, we measured residual concentration of Pb^{2+} after adding BP-9 into PbI_2 aqueous solution (~ 100 ppm). Inductively coupled plasma atomic emission spectrometry (ICP-AES) was employed to obtain the precise concentration of Pb^{2+} in deionized water. As shown in Figure 6c, after adding BP-9, the concentration of Pb^{2+} decreased obviously from 105 ppm to ~ 6 ppm, whereas the BP-5 shows much poorer Pb-absorbing capability compared to the BP-9 additives. Hence, combining the advantages of BP additives (BP-5 and BP-9) is necessary for realizing the entire life cycle Pb-related issues management in PSCs. By incorporation of mixed BP-5 and BP-9 additives in PSCs, the issue of Pb^{2+} leakage has been successfully alleviated without sacrificing the photovoltaic performance of PSCs (Figure S16, S17 and Table S2). Moreover, we have now attained the champion device with 25.47% efficiency through using the high transmittance FTO and continuous experimental optimization (Figure S18 and Figure S19). Finally, to simulate the Pb^{2+} leakage from broken devices under extreme weather, we tested the concentration of Pb^{2+} by immersing the unencapsulated PSCs (mixed BP-5 and BP-9) in 55 ml deionized water. After 20 min, the precise

concentration of Pb^{2+} in deionized water has been obtained by ICP-AES. As shown in Figure 6d, compared to control devices, the target devices exhibited significantly lower Pb^{2+} concentration and the Pb^{2+} concentration in water was dropped by 93.5%. Generally, the dissolution of Pb^{2+} is initial at grain boundaries, as Pb^{2+} within the bulk is temporarily shielded by the outer shell. Moreover, the crystallization process of perovskite tends to push the additives to the grain boundary, because the molecular size of additive is much larger than all the ions in perovskite⁴⁵. Thus, the interaction between BP additives and the Pb^{2+} most likely will take place at grain boundaries, resulting in a layer of water-insoluble composite. Therefore, the leakage of Pb^{2+} at grain boundaries can be alleviated by BP additives. Indeed, since the concentration of BP additives in PSCs is much lower compared to Pb^{2+} , a small amount of additive cannot indefinitely prevent the dissolution of Pb^{2+} . Over time, as the device remains immersed in water, the perovskite film will eventually start to be dissolved. However, more importantly, since the time for start point of Pb^{2+} leakage has been greatly postponed by BP additives, the BP additives provide an important time window for Pb^{2+} leakage warning.

Through Pb-related-issues-management covering the entire life cycles of PSCs *via* BP additives, the highly efficient, ultra-stable and environmentally-friendly PSCs have been successfully developed. Moreover, to further show the greatly improved performance of PSCs by our strategy, we have compared photovoltaic performance of our devices with previous related works in last three years and summarized in Table S3. It is very clear and obvious from the results mentioned in the Table S3 that our device exhibited the best photovoltaic performance, especially the PCE, which further demonstrated the effectiveness of our strategy.

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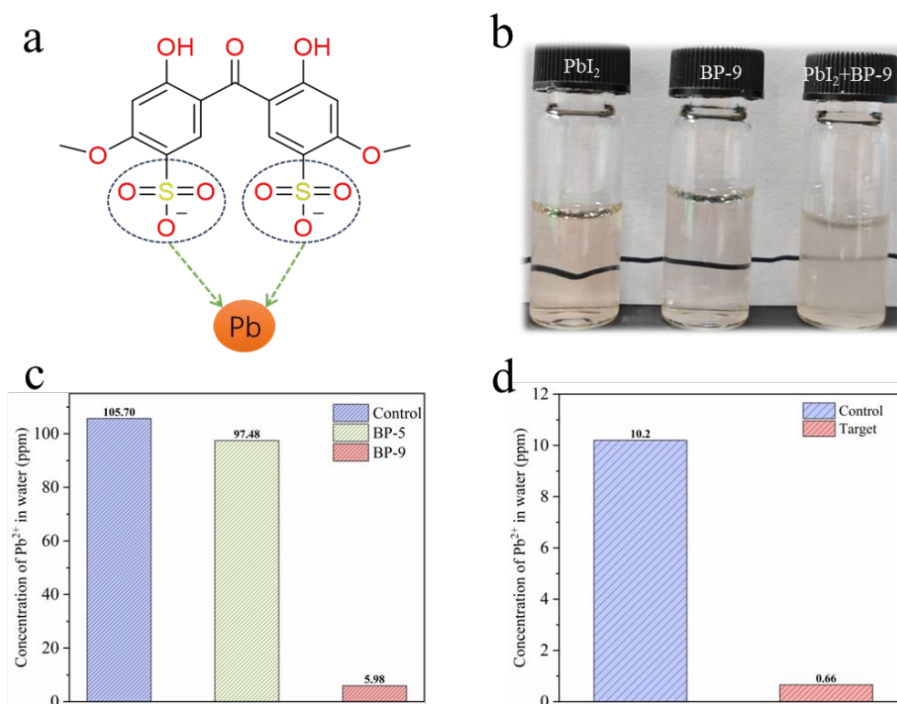


Figure 6. The Pb^{2+} leakage characterization. (a) The chemical structure of BP-9 and the interaction with Pb^{2+} via double sites. (b) The aqueous solution of PbI_2 , BP-9 and mixture (PbI_2 and BP-9). (c) The plot of Pb^{2+} concentration tested by added BP-5 and BP-9 powder into PbI_2 aqueous solution (~ 100 ppm) to prove its Pb absorbing capabilities. (d) The plot of Pb^{2+} concentration tested by soaking two pieces of the unencapsulated PSCs with an area of 1.5×1.5 cm² into 55 mL deionized water for 20 min.

Conclusions

In summary, we, for the first time, proposed a Pb management strategy for systematically alleviating the Pb-related issues covering the entire life-cycle of PSCs, including passivating Pb-related deep-level defects, suppressing the degradation of PbI_2 under the stimuli of UV light as well as the inhibition of Pb^{2+} leakage. Consequently, the resulting champion device yielded a PCE of 25.47% and exhibited high EQE_{EL} of 17.88%. Theoretical and experimental results have revealed that such excellent

performance of target device is mainly due to the effective suppression of the formation of δ -FA phase and passivation of Pb-related deep-level defects. Moreover, owing to the sustainable absorption of UV light of BP molecules, target PSCs have exhibited excellent UV stability that the devices maintained 96% of initial efficiency after ~620h of continuous operation under UV light (22 mW/cm²). While the control devices decayed to ~70% of the initial PCE only after 120 h of aging test, which proves the effectiveness of the current approach. Moreover, due to the passivation of defects and inhibition of δ -FA phase, the operational stability of target devices has been significantly enhanced. Finally, by combining the advantages of BP-5 and BP-9, the issue of Pb leakage has also been effectively alleviated without sacrificing the photovoltaic performance. Through Pb-related-issues-management covering the entire life cycles of PSCs *via* BP additives, the highly efficient, ultra-stable and environmentally-friendly PSCs have been successfully developed, which paves the path for the successful commercialization of PSCs.

Experimental

Materials

FTO (A11-T02) was purchased from Shang Yang Technology Corp. Dimethyl sulfoxide (DMSO), dimethylformamide (DMF), chlorobenzene, isopropyl alcohol (IPA), lithium Bis (trifluoromethanesulfonyl)imide salt (Li-TFSI), and 4-tert-butylpyridine (tBP) were purchased from Sigma Aldrich. SnO₂ colloid precursor (tin (iv) oxide, 15% in H₂O colloidal dispersion) was purchased from Alfa Aesar. Dimethylformamide 2, 2', 7, 7' Tetrakis (N, N-di-p-methoxyphenylamino)-9, 9'-spirobifluorene (Spiro-MeTAD), Tris(2-(1H-pyrazol-1-yl)-4-tert-butylpyridine) and Co (III) TFSI were purchased from Xi'an Polymer Light Technology Corp. Lead iodide (PbI₂) was purchased from TCI. Formamidium iodide (FAI), methylammonium iodide (MAI), Methylammonium chloride (MACl) were purchased from Greatcells.

Device fabrication

The FTO substrates were sequentially washed with ethyl alcohol, acetone, DI water followed by N₂ blow dry. The SnO₂ spin-coated solution was prepared by dissolving 2.67% of SnO₂ nanoparticles into the DI water under constant stirring for 5min at room temperature. Then SnO₂ aqueous solution was spin-coated at 3,000 r.p.m. for 30 s and then annealed in vacuum at 80 °C for 30 min. It was then treated with UV Ozone for 15min to improve hydrophilicity of interface. The perovskite precursor was prepared from 1.5M PbI₂ in DMF: DMSO =9:1 solution and mixed organic salts (FAI: MAI: MACl =90 mg: 6.39 mg: 9 mg) in 1 ml IPA, respectively. Further, 6mg CHMAI was dissolved in 1 ml IPA and stirred at room temperature for 1 hours. The BP molecules were first dissolved in DMF and then added into the PbI₂ solution with various ratios. A two-steps spin-coating method was adopted to deposit perovskite layer. The PbI₂ precursors with different molar ratios of BD molecules were spin-coated at 1500 r.p.m for 30 s onto the SnO₂ and then annealed at 70°C for 1min. Then, the organic salts solution was spin coated onto PbI₂ at 2200 rpm for 30 s, followed by thermal annealing at 150 °C for 15 min in ambient atmospheric conditions (30–40% humidity). Then,

CHMAI solution was spin coated at 5000 r.p.m for 30s onto the perovskite to passivate the interface defects. Afterwards, the hole transporting layer (HTL) was coated onto the perovskite film at 3,000 r.p.m. for 30 s with the dynamic spin-coating process. The HTL solution consists of 72.3 mg of Spiro-OMeTAD, 29 μL of tBP, 18 μL of Co (III) TFSI solution (300mg ml^{-1} in acetonitrile), 25 μL of Li-TFSI solution (520 mg ml^{-1} in acetonitrile), and 1 mL of chlorobenzene. The HTL solution preparation and deposition were performed inside a nitrogen glovebox. Finally, the Au back electrode (85nm) was deposited by thermal evaporation.

Material Characterization.

X-ray photoelectron spectroscopy (XPS) was performed by PHI 5000 VersaProbe III with a monochromatic Al $K\alpha$ X-ray source with the beam size of 1400 μm x 100 μm . Charge compensation was achieved by the dual beam charge neutralization and the binding energy was corrected by setting the binding energy of the hydrocarbon C 1s feature to 284.8 eV. The X-ray diffraction (XRD) measurements were employed by using X-ray diffractometer (D8 Advance) with a monochromatic Cu- $K\alpha$ ($\lambda = 1.5405$ Å) X-ray source. The absorption spectra were obtained between 300 and 600 nm wavelength range using F20-UV thin-film analyzer (FILMETRICS). Synchrotron-based grazing incidence wide-angle X-ray scattering (GIWAXS) measurements were performed at the BL14B1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF) using X-rays with a wavelength of 1.2398 Å. Two-dimensional (2D) GIWAXS patterns were acquired by a Platas2M detector mounted vertically at a distance of ~ 335.94 mm from the sample with a grazing incidence angle ranging from 0.1° to 0.6° . Scanning electron microscope (SEM) images of perovskite films were taken by Hitachi S-4800 and ZEISS SIGMA HD. The micro 2D PL mapping images were tested by a Raman spectrometer (WITEC Alpha300R) under 532 nm laser excitation.

Devices Characterization.

The J - V curves of the PSCs were measured using a Keithley 2602B source in the N_2 filled glove box at room temperature under AM 1.5 G condition at an intensity of 100 mW/cm^2 , calibrated by a standard Si solar cell (PVM937, Newport). The standard Si is calibrated every month. A 450-watt xenon lamp (Oriel solar simulator, 94023A) was used as a light source. The active area of PSCs is 0.105 cm^2 defined by the cross of patterned Au and ITO electrode, and further calibrated by the microscope. The target device was validated by an independent solar cell-accredited laboratory (SIMIT, Shanghai, China) for certification and the aperture area is 0.0638 cm^2 . The J - V curves were obtained both at forward scan (from -0.2 V to 1.20 V, step 0.02 V) and reverse scan (from 1.20 V to -0.2 V, step 0.02 V) without any pre-conditioning before the test. The delay time was set to be 0.05s. EQE data were acquired by an EQE system (Enli Tech, Taiwan) using 100 Hz chopped monochromatic light (300nm to 900nm). Maximum power point dynamic tracking test adopts disturbance observation tracking. The wavelength of UV source is 365 nm and power density of light source is $\sim 22 mW/cm^2$. For the device operating as an LED, the device was mounted in an integrating sphere and the power output was measured for calculation of the EQE (Enlitech). Mott-Schottky curves with capacitance-voltage measurements were performed by a ZAHNER PP211 electrochemical workstation at 10 kHz with bias voltages ranging from 1.2 to 0 V and an AC voltage of 20 mV was used to test the corresponding capacitance at shifty bias voltage. The built-in potential (V_{bi}) in the perovskite layers can be estimated by following depletion approximation equation (Mott-Schottky).²¹ Operational stability was tested by using white LED at 100 mW/cm^2 . Samples were tested under illumination in an N_2 environment at $30 \pm 3^\circ C$ without any encapsulation. Lead Leakage Measurements: the unencapsulated PSCs with an area of $1.5 \times 1.5 cm^2$ were separately soaked into beakers containing 55 mL deionized water (Figure S14). After 20 mins, 8 mL of water sample was taken from each beaker and was put it into a plastic tube as the test sample. All the samples were analyzed by Inductively coupled plasma atomic emission spectrometry (PE-8000).

DFT calculation.

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All the first-principles calculations are conducted based on density functional theory (DFT) as implemented by the VASP code.⁴⁶ The Perdew–Burke–Ernzerhof (PBE)⁴⁷ exchange correlation functional form of the generalized gradient approximation (GGA) is utilized with a 450 eV cutoff energy for the plane wave basis. The structure of α -FAPbI₃ is the one utilized in our previous work,¹² while the structure of δ -FAPbI₃ used here is the same as that Lu et al. utilized.⁴⁸ All the relaxation and self-consistent calculations of the α and δ primitive cells are executed by using the $5\times 5\times 5$ and $3\times 4\times 4$ Monkhorst-Pack k-point meshes,⁴⁹ respectively. The surface structure of α -FAPbI₃ is built based on $2\times 2\times 5$ supercell (240 atoms) of the relaxed 12-atom primitive cell and the surface is (001). The surface structure of δ -FAPbI₃ is built based on $2\times 2\times 2$ supercell (192 atoms) of the relaxed 24-atom primitive cell and the surface is also (001). The surface structures of α - and δ -FAPbI₃ are initially optimized by both using a $3\times 3\times 1$ Monkhorst-Pack k-point mesh. Subsequently, the molecular additives are added to the surface of α - and δ -FAPbI₃ in the same way, I⁻ and FA⁺ on the surface are replaced by C₁₅H₁₁O₅S⁻ and Na⁺, respectively. The method of replacement derives from the previous work.^{26, 50} All the relaxation and self-consistent calculations of the related surface structures with molecular additives are executed by both using a $3\times 3\times 1$ Monkhorst-Pack k-point mesh. The convergence criteria of the primitive cell relaxation is 0.01 eV/Å, while that for surface structure relaxation is 0.05 eV/Å. To remove the interaction between the mirror images of slab models along [001] direction, a vacuum layer larger than 15 Å is used to isolate the slab models.

Author Contributions

K.L., conceived the idea, Y.Z., A.Y. directed and supervised the project. K.L. initialized this project and fabricated the devices. T.H. and F.L. took part in experiment test. Z.C. and S.C. performed DFT simulation. K.L. wrote the first draft of the paper. S.R, A.Y. and Y.Z. modified the paper. All authors contributed to discussions and to finalizing the manuscript.

Competing interests

The authors declare no competing interests.

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